asked to swallow a contradiction: on the one hand, they're told that the idea of a classical object is logically prior to the idea of quantum physics, since the idea of a classical object is needed in order to figure out when a measurement has happened; but, on the other hand, they're told that quantum physics is logically prior to classical physics, the latter emerging out of the former. These ideas can't both be right. And, in practice, the version of the Copenhagen interpretation most commonly found in textbooks and "in the wild" values the first idea over the second. Some objects are just classical, and it's interaction with those objects that is defined as measurement for the purposes of quantum physics, "solving" the measurement problem well enough to allow calculations to be performed. Certainly most physicists (including yours truly) also believe that quantum physics underpins classical physics, but when actually performing quantum physics calculations, this fact is conveniently forgotten, and some objects are simply treated as exempt from the Schrödinger equation. Hence the desperate need to shut up while calculating.

There have been physicists who have tried to bring the idea that quantum physics is fundamental into their calculations. To do so, they had to give up Copenhagen's solution to the measurement problem and perform conceptual work to solve the problem a different way. In other words, these people, people like David Bohm and Hugh Everett, had to develop new interpretations of quantum physics, because the Copenhagen interpretation doesn't take quantum physics seriously. It requires that we abandon the idea that quantum physics can be used to deal with everything in the universe and restricts it to a limited domain. Today, most physicists agree with Zeilinger that there is no limit to the validity of quantum physics—but the manner in which quantum physics is generally taught and used betrays that ideal.

Nonetheless, the appeal of Copenhagen makes some sense, seen in this light. Quantum physics drove much of the technological and scientific progress of the past ninety years: nuclear power, modern computers, the Internet. Quantum-driven medical imaging changed the face of health care; quantum imaging techniques at smaller scales have revolutionized biology and kicked off the entirely new field of molecular genetics. The list goes on. Make some kind of personal peace with Copenhagen, and contribute to this amazing revolution in science… or take quantum physics seriously, and